

Sealed Experiment Record

Sealed 2026-10-07 · Signed by the K-Veritas server
Session 501cbbb0 · Data hash 7955...ff80

Abstract

This record seals 1 run (kv_run.sh) executed on Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz between 2026-10-07 15:49 and 15:49 UTC. Reported results: training_seconds 15.66, compression_basic 2.35908, compression_regex 2.12698, tests_passed 21. 1 invocation in total, 0 failed. Produced with K-Veritas [1].

What this record proves

These numbers came from the sealed code at this time, unchanged since. Whether the experiment is sound is for the reader to judge.

Verify: kveritas verify <this file> or kveritas.org/verify

1 Setup

Item	Value
Session	501cbbb0-1213-4c15-b180-254a42166ab6
Initialized	2026-10-07T15:49:20Z
Machine	167ec0d6dc9a8170
Operating system	linux / amd64
CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
Memory	31.0 GB
GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
Environment, run 1	2f80f9b4d22d6e54a6e0f3cd60a1ee15eb6edd4f0e961930ff15480e6a1eeb49
Disclosure	open
Source code, run 1	ff1caa598396d197ce6aab5d05135cdaac3c9de61364b759c77ce0fffc087dd6

2 Runs

#	Command	Duration	Exit	Anchored
1	bash kv_run.sh	32.8 seconds	0	at run end

1 invocation in total: 1 kept in this record, 0 failed, 0 interrupted. Every invocation is recorded by the server, including the ones not kept.

3 Results

Run 1: kv_run.sh

Metric	Final value	Step	Source
training_seconds	15.66		explicit
merges_logged	512		explicit
compression_basic	2.35908		explicit
compression_regex	2.12698		explicit

Metric	Final value	Step	Source
tests_passed	21		explicit
tests_failed	0		explicit

Inline claim	Value	Phase	Output line
training_seconds	15.66		517
compression_basic	2.35908		520
compression_regex	2.12698		522
tests_passed	21		528

4 Execution evidence

Coherence

Coherent (0.90): the telemetry channels move together, as one process.

Telemetry

Run	Samples	CPU time	Peak memory	Peak GPU
1	313	15.7 s	0.12 GB	0%

5 Provenance

Run 1: kv_run.sh

Disclosure open. 9 tracked files, 5 snapshots. Every snapshot is bound into the signature.

#	Snapshot	Time	Root	Changed
0	run_start	15:49:20	e199fd5a5b300ecb	9
4	run_end	15:49:53	e199fd5a5b300ecb	0

3 snapshots with no file change omitted here; full timeline in Appendix B.

Activity: 11 files read, 6 files written, 5 subprocesses. Full list in Appendix B.

6 Verification

Field	Value
Signer	Signed by the K-Veritas server
Signed at	2026-10-07T15:49:53.689345+00:00
Algorithm	RSA-PSS-SHA256, 4096-bit key
Key fingerprint	eeac81653cf44ac029c4069cf2a06d9f
Data hash	7955540c833ac615d223dba88b38b2a56fdab743ca0f614c9d6a92ec5a5dff80
Code bundle hash	c99f4be5dff693676a71fd28906bf6b185e89d896fe034894227db3a6aa4a964

Run anchors: 1 invocations were anchored at the server when they ended.

Checks a verifier runs

1. The stored canonical data hashes to the data hash above, and the signature verifies over data_hash:nonce:signed_at.
2. The signing key is the K-Veritas server key, not one supplied by the author.
3. Nothing was appended after the seal, and these pages hash to the value bound in the seal.
4. Each run matches the digest the server received when it ended.
5. A supplied code bundle hashes to the code bundle hash above.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

Appendix

A Full metric series

Run 1: kv_run.sh

Metric	Value	Step	Line	Source
training_seconds	15.66		515	explicit
merges_logged	512		516	explicit
compression_basic	2.35908		519	explicit
compression_regex	2.12698		521	explicit
tests_passed	21		526	explicit
tests_failed	0		527	explicit

Phase	Line	Time	CPU time	Memory	GPU
train	1	15:49:20	0.4 s	0.03 GB	
compression	518	15:49:36	0.0 s	0.03 GB	
tests	523	15:49:42	0.1 s	0.02 GB	

B Provenance timeline and activity map

Run 1: kv_run.sh

#	Snapshot	Time	Root	Changed
0	run_start	15:49:20	e199fd5a5b300ecb	9
1	phase train	15:49:20	e199fd5a5b300ecb	0
2	phase compression	15:49:36	e199fd5a5b300ecb	0
3	phase tests	15:49:42	e199fd5a5b300ecb	0
4	run_end	15:49:53	e199fd5a5b300ecb	0

Event	Path or command	Hash
read	kv_run.sh	
read	train.py	
read	minbpe/__init__.py	
read	minbpe/basic.py	
read	minbpe/gpt4.py	
read	tests/taylorswift.txt	
read	tests/__init__.py	
read	tests/test_tokenizer.py	
write	models/basic.model	ecff325f8c514737
write	models/basic.vocab	8101c1b0dc42b9a5
write	models/regex.model	b58d827c48c4403a
write	models/regex.vocab	1888ba58e7b8571e
read	minbpe/base.py	
read	minbpe/regex.py	
write	test_tokenizer_tmp.model	
write	test_tokenizer_tmp.vocab	
read	.gitignore	
process	tee ../train.out	
process	../env/bin/python train.py	
process	../env/bin/python -	

Event	Path or command	Hash
process	tee ../tests.out	
process	../env/bin/python -m pytest -q tests	

C Run history

Invocation 1, OK, 32.8 seconds, 528 output lines, started 2026-10-07T15:49:20

metric_hash fed4a50341fdc7b9dbc7a23bde65add8727a761f4f9a7b50d1d6590a57e12bf1
run_digest c80555ae211c4ab589b3a364475f68d55ac1dddda21fb987eae7a15c19d1241
anchored at run end

D Raw seal fields and hashes

Seal

data_hash 7955540c833ac615d223dba88b38b2a56fdab743ca0f614c9d6a92ec5a5dff80
signed_message_hash 9e44e67242c3fb222ee3600cf71fb9c80a07f066714eea3949680d5c9ecfdc4d
nonce 0e2608e8304498a7b2273dfd701dbcfaf
signed_at 2026-10-07T15:49:53.689345+00:00
sealed_at 2026-10-07T15:49:53.749713517Z
checkout_bundle c99f4be5dff693676a71fd28906bf6b185e89d896fe034894227db3a6aa4a964

Signature (base64)

nYw9+mtotIsf2eul7TenL797VYbpdnXBG89Ubr4uXSorywigASnPgbfGXRnqkfMQ0+1Lk1wWSupqIbEIL+x35+e
Uo1iYiquB1bUbymA1KQMuzF7JyguI6Ku3w1Sfpl0lqQo5Eu64EChvtNeVpnwidNhyfgu7tYs6JWxbYd5XfpvzbIF
UyXRgs8th2SqW5Jzfa/vpU5ItI21FvJS2M/VRfKwWCbnqBbt2Nc6RX07S9SuGdDcv0Za/5DkU/SapV90ZvX9QiPb
l7rkYf7+cu/PrJYxBqsd0K4/oHiH8o4TQoA0NhBwIFir4nazdwFF6/F1CDIUyq+L01B4aqUheihNXQaoWWPUXEDN
svNC9zxUNPE3Kd3L0uTAsBMQnLqWxydA/S2HncCJr9KxSo9nNltdB/whYo859cNKx6M2atuhiDvSbm8iAFVyqDwf
Uex7WV0oxvI4BQIYe8pyXEBRYDs0d6TZT2g9aFy0NaqJUw6Q2/DYC/RbX/IVzKmXCQRtG+teyCaPiwg8ifoIfa0t
6e+6CW0lFuapiPYK2vADPdmC/19b0EOVhKgmRCljZAC2jwhwMZ0kSpkv9BqvBd9vcyv50EEIbctmwVy0l2pUVQ/u
gG3g6yfQjgheBEeG7vD3HThrP5x0J/CPf6tnUkbHF6ebJwllHbQjnhD0xYp4yALurDk=

Public key

-----BEGIN PUBLIC KEY-----
MIICijANBgkqhkiG9w0BAQEFAAACAg8AMIICCgKCAgEA1QL3YyMgeShx8IhTC7jb
sM0w2BnRXgsBn3+SMXKjtwHATqdekmlPlcjayJxD5fcProPPCqp/LouFz5CUZzRi
6+8FTv70dHuIz5A4Lat4vcgwP+iWscPc3HzpmjpsKbTmOemqgp6RB6GNNpxYa5X
7qU95EFeisX/GfoS9ToxtqPBwCcVWSPBX8efec4TC58m96Fsp4mC8KICeadbIbHN
av0JjdTJ//CM0WZSIbdzE6qvEbNFCqDB5tNU7oHfYvBeNlrvPU9TAqdbtozdHq+D
AhfTRuJNgWEKjwD5lb0Vhi4qlkweSOEMTKT3ye9I15xEaJn4bku4eKqes7VSzt3o
XbUcmr2H5mKcKsvt6bXmX6ajAmgi900Cm9QJlaQkJM6z2iITnyvDMgueJtFxQpC9
HNBPgPZ242NH8JgE6CNF7ToL9syas67u7UMgICnGeNHNCy9bLgPU5annzP9gN2l
6HEMldkXNdKQRb1rDwyEJf3N/50YHnXXPmuf5sb8R/pxWNqtF04R5UGVV92qd2Mi
1GN7lE3mbQRbPsLL/JAPDXMNZNikaFfBsQK7iKxr7vIaTeZckFZWfThcYNT7Hasg
+eE4bf/olCHBCf1VIDCkDxohehdZ0WRRcgM9iK3I/0zZdaBsgdc8WuJRmb8LPP9v
8Jap/1PWjNcNku5cdW3gHLcCAwEAAQ==
-----END PUBLIC KEY-----

Run 1 outputs

stdout bacb432a180bfeb2543859aa96b03aca96453bb34b0b1fabe1ded5bc217855eb
stderr e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855
metrics fed4a50341fdc7b9dbc7a23bde65add8727a761f4f9a7b50d1d6590a57e12bf1
digest c80555ae211c4ab589b3a364475f68d55ac1dddda21fb987eae7a15c19d1241
kv_run.sh (unchanged)
pre 82b25afa5cdc259141ded8cbb170823c0c135631e2e8f7e6633d8e948ac0211a
post 82b25afa5cdc259141ded8cbb170823c0c135631e2e8f7e6633d8e948ac0211a

E BibTeX for [1]

```
@misc{keita2026computerscienceconferencesrequire,  
  title={Computer Science Conferences Should Require Nonrepudiable Experimental Results},  
  author={Mamadou K. Keita and Christopher Homan},  
  year={2026},  
  eprint={2605.08586},  
  archivePrefix={arXiv},  
  primaryClass={cs.CR},  
  url={https://arxiv.org/abs/2605.08586},  
}
```